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Sky LIS Prototype Comparison

Old versions (Client Portal v1, Super Admin v2) against the new versions (Client Portal v2, Super Admin v3): enhanced pages, new pages and new modules with their descriptions and dependencies

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Compared	Client Portal v1 → v2 · Super Admin Portal v2 → v3		
Sources	Sky-LIS-Client-Portal-Interactive-Prototype-v1.html · Sky-LIS-Super-Admin-Production-Prototype-v2 (1).html · prototypes/dist/Sky-LIS-Client-Portal-Prototype-v2.html · prototypes/dist/Sky-LIS-Super-Admin-Portal-Prototype-v3.html · docs/knowledge/sky-lis/SkyLIS_vs_LDM_Gap_Analysis.md · docs/requirements/LIS_Requirements_Specification.html		

32	38	9	17	18	5
Client pages enhanced	Client pages added	Client modules added	Admin pages enhanced	Admin pages added	Admin modules added

> Reading guide: "Enhancements" lists what changed on a page that already existed in the old prototype. "Dependencies" names the master data, other pages or modules a page needs in order to work (what must be set up first, or what it reads from). "—" means the page depends only on itself.

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1. Cross-cutting enhancements (both portals)

Area	What changed	Dependencies
Brand & design system	Logo embedded on sidebar, login, public page and every printed document; palette derived from the logo (blue → violet gradient for primary actions and active navigation, navy for headings and sidebar, specimen red for critical/danger). One shared CSS design system for both portals.	prototypes/src/core.css brand/sky-lis-logo.webp
Bilingual shell	English / Arabic with true RTL on both portals (the old Super Admin was English only); all master data carries Arabic names; statuses, flags and messages translated.	i18n dictionary (≈ 750 keys)
Navigation shell	Grouped sidebar with menu filter, counters (pending treasury requests, delayed tests, verified-unprinted reports), collapsible navigation, branch pill, wallet pill, notifications, profile menu, keyboard shortcuts (Ctrl+S save, Ctrl+K search, F2 service picker, Esc).	core.js shell
Generic grids & editors	Every list has search, status and type filters, Excel export, pagination, row actions; every editor validates mandatory fields, blocks deletion of referenced records and offers deactivate-instead-of-delete.	core.js CRUD framework
Printed documents	Barcode labels, job order, receipt, invoice, refund receipt, price quote, laboratory report (templates, verifier signature, QR, reference ranges, flags, comments), abbreviated worksheet, worklist, cash-closing sheet, claim, analyser code sheet.	Report Templates Users (signature image) Parameters (auto-print)
Audit trail & notifications	Every registration, payment, refund, transfer, cancellation, status change, print and privilege change is written to the audit log with user, time, branch and device; notification centre for panic values, delays, treasury requests and agent status.	Audit Log page notification centre
Demo data model	Patients, visits with order lines, samples with status events, results with logs, payments/refunds/transfers, payers with tiers, base and contract price lists, analysers with mappings, PCR kits, verification settings and medical rules — enough to exercise every workflow end to end.	data.js

2. Client Portal — v1 to v2

2.1 Pages that existed and were enhanced

Old page	New page	Module	Enhancements	Dependencies
Public website	**Public website**	Access	• Rebranded with the logo and palette • Six feature cards now cover sample lifecycle, analyser integration and auto-verification • Live KPI card uses the demo tenant	Brand assets, i18n
Subscription wizard	**Subscription wizard**	Access	• Same four steps (lab details → OTP → credentials → portal URL) restyled • Mobile length validated against the country master • Arabic lab name captured	Countries & governorates (Super Admin master data), Notification templates (OTP, portal URL)
Login / forgot password / branch chooser	**Login / forgot password / branch chooser**	Access	• Logo and bilingual toggle on the login page • Branch chooser explains that only assigned branches are listed and every transaction is branch-stamped • OTP reset flow leads to the password-policy form	Users (branches), Branches
Dashboard	**Dashboard**	Overview	• KPIs extended from 5 to 8: samples received, pending results, unverified backlog, delayed tests, rejected samples, wallet • Delayed-tests table computed from turnaround settings • Recent activity fed by the audit log • Quick actions cover collection, work queue and cash closing	Worklist engine (durations), Audit log, Results & samples data

Old page	New page	Module	Enhancements	Dependencies
Registration	**Registration**	Reception	• Patient file: auto patient number, title → gender, age ↔ date of birth (both directions), patient condition, national ID, nationality, governorate/city, reception comments (CR), VIP flag • Duplicate protection: similar-name modal with number and mobile, patient search with % wildcards, "use this file" vs "new file" • Visit: location type (out/in-patient/emergency), department, room, treatment number, referring doctor, payer + contract tier, insurance ID, diagnosis (adds its service set), urgent / biohazard / collected-outside flags • Service ordering: search by name, alias, short code, shortcut number or shortcut set (Enter), picker (F2) with group filter, precautions highlighted, gender restriction, frequency check (alert or prevent), contract alert pop-up, auto-add rules, duplicate guard • Per line: Ordered/Reserved status, expected result date, patient share and payer share from the contract, line discount, collected-by • Preparation questions linked to the selected services with Yes/No, option list and text answers; "No" suggests Reserved • Summary: payer vs patient share, discount by amount or percent with per-user cap and mandatory reason, live preview of the tubes that will be generated (group + sample type + condition rule) • Save: creates accession, samples with status events, results per test; applies auto-collect/auto-receive	Services & Tests (sample, condition, duration, precaution) Payers & Contracts + Contract Pricing Patient Conditions Preparations Diagnoses Shortcut Sets Test Rules Doctors Reasons (discount) Parameters (mandatory fields, auto-print, auto-receive, similar-name, discount cap) Treasuries (payment step)

Old page	New page	Module	Enhancements	Dependencies
			parameters; success modal with labels, job order and payment; Ctrl+S • Edit mode for a saved visit: add services (new tubes), reactivate reserved lines, update demographics	
Search Registration	**Search registrations**	Reception	• Filters by accession, patient number, name, mobile, date range, payer and status • Row shows batch number, urgent flag, service chips with reserved/cancelled styling, patient share, remaining balance badge • Actions: edit visit, samples, payments, documents menu (receipt, job order, labels, report), open results	Visits data · Documents · Results by Accession
Patients	**Patients**	Reception	• Search by number, name (Arabic too), mobile or national ID • Patient condition and VIP shown; visit count and last visit computed • Visit history modal with results links • Patient documents (attach, tenant-wide) • Merge duplicate patients (audited) • Start a new visit from the file	Visits & results · Audit log Special right: Merge Patients
Search Payments	**Payments**	Finance	• Shows total, discount, patient share, paid, refunded and remaining per visit • Multiple partial payments; methods cash, card, POS, bank transfer with mandatory reference • Each payment tied to treasury, cashier and device; receipt printed automatically when the parameter is on • Receipts history with reprint and invoice	Treasuries & Treasury Requests (device approval) Parameters (auto-print receipt) Report Templates (receipt)

Old page	New page	Module	Enhancements	Dependencies
Query Service	**Price query**	Reception	• Quote per payer and tier with patient share, payer share, list price, expected result date and preparation • Services added by shortcut number, short name or picker • Printable quote; "Continue to registration" carries the services and contract into a new registration	Contract Pricing Services (duration, precaution)
Worklist	**Worklist**	Laboratory	• One row per test (not per service) filtered by collection date • Filters: group, test, analyser, sample status, payer; boxes for new, remaining, completed, manual, not verified, not printed, incomplete, printed, delayed • Counters: requests, tests, samples • Expected result date from duration, delay computed, row colour by percentage of duration consumed (tenant thresholds) • Sample tracking from the row; printable bench worklist	Tests (duration, min duration, test days) Samples (collection events) Analyser Tests (manual vs interfaced) Parameters (delay thresholds)

Old page	New page	Module	Enhancements	Dependencies
Results	**Results by accession**	Laboratory	• Opens in query mode: accession, patient number, name with %, test status, sample status, from-analyser, not-printed / not-verified / delta filters; matching visits listed with colour chips per report unit • Case header: age, sex, condition, location, doctor; CR (reception comment, red when present), preparations, diagnosis, VIP / urgent / biohazard, previous-results marker • One card per report unit (visit × clinical group, print-alone services separate) with status colour and received state • Grid: result input typed by test type (numeric, list, optional list, formula-locked, culture, text), unit, resolved reference range (sex / age / condition), flag incl. panic, analyser source, sample status, test status, delta flag, printed flag, attachments, result log, status log • Three comment kinds: printed report comment with library insert, reception CR, internal notes • Actions: Save (Received gate, Restricted-list validation, formula recalculation, delta evaluation, panic alert), Review, Verify (rights per group, delta interception, review-first parameter), Undo, Preview (refused while values missing), Print (verified only) • Culture entry (organisms, antibiotics, sensitivities, no-growth) and formatted-text entry (paragraph headings, templates, spell check, voice note) • Report modal with template, signature, QR, ranges and flags; reportable limits printed as < or >	Tests (ranges, values, formulas, delta settings) Patient Conditions Comment Library Roles (S/R/V/E/P per group) Samples (Received gate) Microbiology masters (cultures) Report Templates & Users (signature) Parameters (review-first, print unverified) Previous results

Old page	New page	Module	Enhancements	Dependencies
Roles	**Roles & privileges**	Setup	• Screen privileges as Query / Insert / Update / Delete per screen (instead of six module toggles) with filter and grant/revoke all • Special rights (discount with cap, payment, refund, transfer, cancel, modify verified result, preview without verify, print invoice, VIP…) • Result rights See / Review / Verify / Edit / Print per clinical group and per branch • Allowed patient locations • Validation: active role needs at least one screen; role with users cannot be deleted	Clinical Groups · Branches · Screen registry
Users	**Users**	Setup	• Arabic name, designation, role, branches (multi), shifts • Security tab: password policy, expiry date, must-change flag, lock/unlock, signature image upload • Privilege overrides with "modified" flag and restore-to-role • Main administrator protected	Roles · Branches · Report Templates (signature use)
Branches	**Branches**	Setup	• Country and governorate from master data, currency, email; deletion blocked when visits exist	Countries & Governorates
Clinical Groups	**Clinical groups**	Setup	• Unit code printed on labels, report template per group, display order; deletion blocked when services exist	Report Templates

Old page	New page	Module	Enhancements	Dependencies
Services	**Services**	Setup	• Types extended to Test / Package / Culture / Text / Additional • Sample type and condition, header, aliases, precaution, gender restriction, duration, label count, print-alone, visible at ordering, visible in job order • Members with order, "in sum" and visibility; profile sum; packages contain services • Base price shown; filters by type and group	Tests, Sample Types & Conditions, Clinical Groups, Base Price Lists
Tests	**Tests**	Setup	• Four tabs: information, configuration, timing & alerts, labels & print • Numeric: several reference-range bands by sex, age and patient condition with normal and panic limits; reportable limits; decimals • Lists: values with normal / critical / default / colour and the "no Negative default" rule; optional vs restricted • Formula: expression with validation against test codes • Formatted text: paragraph structure and default text • Timing: duration, minimum duration, frequency check (alert / prevent), delta % and window, test days • Labels count, prefix, report header, print-reference source; visibility and biohazard flags	Clinical Groups, Sample Types & Conditions, Units, Patient Conditions
Service Global Sequence	**Service sequence**	Setup	• Inline sequence with uniqueness validation (unchanged behaviour, new shell)	Services
Sample Types	**Sample types**	Setup	• Sample conditions per type (fasting, random, refrigerated…) — the second half of the tube-merge key	—

Old page	New page	Module	Enhancements	Dependencies
Units of Measurement	**Units of measurement**	Setup	• Deletion blocked when referenced by tests	—
Commercial Drug Types	**Drug types**	Setup	• Unchanged behaviour, new shell	—
Antibiotics	**Antibiotics**	Setup	• Linked to commercial names; used by culture entry	Drug Types
Organisms	**Organisms**	Setup	• Used by culture entry; "No growth" entry	—
Preparations	**Preparations**	Setup	• Option list values (minimum two), gender applicability, linked services, Arabic question; answers captured at registration and shown in results	Services
Comments	**Comment library**	Setup	• Scope per clinical group, Arabic text; inserted into report comments with % search	Clinical Groups
Doctors	**Doctors**	Setup	• Branches (multi), share percentage for doctor-income report, phone	Branches
Treasuries	**Treasuries**	Finance	• Shows approved users per treasury; branch from master	Treasury Requests
Request Treasury	**Treasury requests (request button)**	Finance	• Request is a modal on the requests page: branch, treasury and the current device ID; status Pending	Treasuries
Treasury Requests	**Treasury requests**	Finance	• Approve / reject / revoke with audit entry; approved users feed the treasuries page	Treasuries, Audit log

Old page	New page	Module	Enhancements	Dependencies
Payers	**Payers & contracts**	Finance	• Payer type (cash / credit / insurance / lab), search key, web-results portal credentials • Contract tiers (English | Arabic) maintained on the payer • Jump to contract pricing per payer	Contract Pricing
Pricelists	**Base price lists**	Finance	• Named yearly lists with active flag, inline prices, derived-contract list • Copy a list into another with × factor or + amount • Apply propagates to derived contracts (stamps apply date)	Services, Contract Pricing
Profile	**Profile**	Account	• Portal URL and plan shown; credential change confirmed by OTP	Subscriptions & Wallets (Super Admin)

2.2 New pages

New page	Module	Description	Dependencies
Home reservation	Reception	Books a home visit: patient data, mandatory address, visit time, assigned phlebotomist, services with quote; booking number; preliminary labels; cancel; convert to a registration without retyping.	Services, Payers & Contracts, Users (phlebotomists), Registration
Batch registration	Reception	One company contract, one service set for all members, one row per person with patient-number lookup; creates one visit per member under a batch number and hands over to batch payment.	Payers & Contracts, Services, Doctors, Batch Payment
Lab-to-Lab registration	Reception	Samples arriving from a contracted laboratory: choosing the lab fills contract, courier and phone; per-row patient data and services; "collected inside" responsibility flag; batch number; billing to the sending lab.	Sending Labs, Payers & Contracts, Services, Claims

New page	Module	Description	Dependencies
Print reports	Reception	Preset of the work queue listing verified-unprinted reports with the print list builder, print list numbers and the abbreviated worksheet.	Work Queue, Results
Collection & receipt	Samples	Barcode scan in Collect or Receive mode, whole-accession actions, today's samples with status counts; enforces Ordered → Collected → Received and the Received gate for results.	Samples generated at registration, Parameters (auto-collect / auto-receive)
Samples by visit	Samples	All tubes of a visit with unit, type, condition, status, labels and services; reprint one label or the job order.	Visits, Documents
Reject / unreject	Samples	Reject a tube by barcode with a reason (notifies reception); reinstate under the same barcode or issue a replacement with a new barcode per tenant policy; list of rejected samples.	Reasons (rejection), Parameters (new-barcode policy), Labels
Sample tracking	Samples	Timeline of locations and statuses with users, time between stages and test round times (complete, reviewed, verified, printed).	Samples events, Results status logs
Results by sample	Laboratory	Scan a tube and work only the tests on it, with the same case panel as results by accession.	Results by Accession, Samples
Work queue	Laboratory	Status-driven list of report units: registration-date range, group, payer, from/to analyser, normal/abnormal, received-only, keep-previous; exact status boxes and cumulative "Not X" boxes; nine colour states; bulk review/verify; print list with numbering and abbreviated report; analyser closing presets.	Results, Analysers & Analyser Tests, Roles (R/V rights), Report Templates
Quality control	Laboratory	Placeholder for QC materials, targets, Levey-Jennings and Westgard rules; sample QC runs table. Scope to be confirmed.	Analysers, Tests
Refunds	Finance	Negative balances listed; refund with amount (never above the refundable amount), method and reason; refund receipt for patient signature; audited.	Payments, Reasons (refund), Special right: Refund

New page	Module	Description	Dependencies
Money transfer	Finance	Contract transfer: move a visit to another payer/tier with re-pricing preview and resulting balance (excess becomes a refund). Line transfer: edit actual patient/payer amounts per service. Logged and reported.	Payers & Contracts; Contract Pricing; Refunds; Special right: Money Transfer
Cancellations	Finance	Cancel services of a visit with a reason; verified services blocked; related samples cancelled; receivables adjusted; cancellations report.	Reasons (cancellation); Samples; Special right: Cancel Services
Batch payment	Finance	Settle a batch or lab-to-lab batch by number, member by member or pay-all; lab batches billed through claims.	Batch Registration; Lab-to-Lab Registration; Payments
Cash closing	Finance	Per treasury, cashier and period: payments and refunds listed, expected cash = cash payments − refunds, card/POS separate, outstanding balances and discounts; printable handover sheet.	Payments; Refunds; Treasuries
Claims	Finance	Claim to a credit payer for a period or tier in six layouts (standard, per tier, with results, totals only, with counts, credit note) with totals and approval boxes; drafts and issued claims; Excel export.	Payers & Contracts; Visits with payer shares; Report Templates (claim)
Contract pricing	Finance	Per payer tier: up to three base lists with patient and payer factors, effective dates, Apply with date stamp, derived price grid with per-service exclusion (negotiated price) and alert condition, copy from another tier.	Payers & Contracts; Base Price Lists; Services
Financial reports	Finance	Sixteen reports: patients' income, payment methods, contract type, monthly statistics, patient statement, doctor income with share, clients income, service income analysis, outstanding balances, discounts, cancellations, transfers, total per day, price-list hand-out, cost vs income, turnaround compliance.	Visits, payments, refunds; Doctors (share); Tests (durations)
Patient conditions	Setup	Conditions (pregnant, diabetic…) selected at registration that switch the reference range band.	Tests (range bands)
Diagnoses	Setup	ICD codes with linked service sets proposed at registration.	Services

New page	Module	Description	Dependencies
Sending labs	Setup	Contracted laboratories with payer/tier, courier and contacts for lab-to-lab registration and claims.	Payers & Contracts
Shortcut sets	Setup	Typing shortcuts (ICU, DM, ANC) that drop several services at their own prices.	Services
Test rules	Setup	Auto-add rules: trigger services >> added services, validated for contradictions.	Services
Reasons	Setup	Reason lists for rejection, cancellation, discount and refund.	—
Report templates	Setup	Templates per report type with logo, signature, QR and language.	Clinical Groups (template per group)
Additional services	Setup	Chargeable non-test services (SMS, home visit, urgent processing) without samples or results.	Base Price Lists
Parameters	Setup	Tenant switches: mandatory fields, similar-name check, discount cap, auto-print, print unverified, auto-collect / auto-receive, new-barcode policy, review-before-verify, delay thresholds, session timeout.	—
Analysers	Integration	Analyser definition with mode (uni / bi-directional, download, shared folder), serial or TCP settings, auto-print, download, append-unregistered policies, overwrite policies and a flag table (stop / record); connection test.	Clinical Groups, Sample Types, Site Agent
Analyser tests	Integration	Per-analyser test mapping: analyser code, host code, result selection, slope / intercept, unit, query and query details, Auto Send (save permission), text conversion table, analyser-specific range; printable code sheet.	Analysers, Tests
Analyser samples	Integration	Sample-type codes required by some machines.	Analysers, Sample Types
Site agent	Integration	Per-branch agent status, heartbeat, messages and errors, connected analysers, configuration and the decrypted hourly log; push configuration, fetch logs.	Analysers, Branches

New page	Module	Description	Dependencies
PCR kits	Integration	Kit interpretation: instrument, serial lock, run-file layout, per-gene positive / borderline Ct thresholds, verdict rules, result / backup / failed folders; run simulation with audit flags.	Analysers (PCR instrument)
Verification settings	Automation	Master switch for online verification, per-group checker set (medical rules, no-rules guard, flags, panic, delta, group completeness, inter-analyser delta…), simulation, result options and result dictionary.	Medical Rules, Tests, Analysers (flags)
Medical rules	Automation	Rule editor per test with parameter list, operator keypad, formula condition, recommendation text, active flag and syntax validation; rule-writing lessons.	Tests (LOINC codes, ranges), Verification Settings
Wallet	Account	Prepaid balance, next charge, SMS usage and wallet transactions; top-up request.	Subscriptions & Wallets (Super Admin)
Support	Account	Support tickets with priority and status.	—
Audit log	Account	Tenant audit trail of every action with user, time, branch and device.	All transactional pages

2.3 New modules

New module	Description	Pages	Dependencies
Samples (sample lifecycle)	Tube generation by rule, barcode collection and receipt, the Received gate, rejection and replacement, tracking.	Collection & receipt, Samples by visit, Reject / unreject, Sample tracking	Services & Tests (sample type, condition, labels), Reasons, Parameters, Labels & job order
Laboratory work queues & turnaround	Status-driven queues with Not-X logic, bulk sign-off, print lists, worksheet; worklist by collection date with TAT colours.	Work queue, Print reports, Worklist, Results by sample	Results engine, Tests (durations), Analysers, Roles

New module	Description	Pages	Dependencies
Pricing engine	Base lists, contract tiers derived by formula, exclusions and alerts, patient / payer split on every visit.	Base price lists · Contract pricing · Payers & contracts · Price query	Services · Payers
Cash & claims	Payments bound to treasuries and devices, refunds, transfers, cancellations, batch payment, daily closing, claims and the financial report set.	Payments · Refunds · Money transfer · Cancellations · Batch payment · Cash closing · Claims · Financial reports	Pricing engine · Treasuries & requests · Reasons · Roles (special rights)
Registration modes	Home reservation, batch and lab-to-lab registration beside the walk-in flow.	Home reservation · Batch registration · Lab-to-Lab registration	Registration · Sending labs · Batch payment
Integration	Analyser definitions, code mapping, sample codes, site agent and PCR kit interpretation.	Analysers · Analyser tests · Analyser samples · Site agent · PCR kits	Tests · Clinical groups · Results engine (analyser column, From/To analyser)
Automation (auto-verification)	Engine settings, checkers, result options/dictionary and the medical-rules language.	Verification settings · Medical rules	Tests · Analysers (flags) · Results engine
Catalogue depth	Patient conditions, diagnoses, sending labs, shortcut sets, test rules, reasons, report templates, additional services, parameters.	Patient conditions · Diagnoses · Sending labs · Shortcut sets · Test rules · Reasons · Report templates · Additional services · Parameters	Services & Tests
Account & governance	Wallet, support and the tenant audit log.	Wallet · Support · Audit log	Super Admin subscriptions · All pages (audit)

3. Super Admin Portal — v2 to v3

3.1 Pages that existed and were enhanced

Old page	New page	Module	Enhancements	Dependencies
Login	**Login**	Access	• Logo and bilingual toggle; same username/password flow	Brand assets
Home	**Home**	Overview	• Logo card with welcome line	—
Dashboard	**Dashboard**	Overview	• Period selector kept; added wallets-at-zero KPI, subscription vs manual split and plan distribution; recent clients with URL, plan and wallet	Clients Subscriptions & Wallets
Clients	**Clients**	Administration	• Plan, wallet, portal URL, licence, branches and users counts • Row actions: reset password, open wallet, activate / deactivate (deactivation blocks all client users) • Creating a client generates its master-data copy; filters by source and plan	Countries & Governorates Plans Master data (copied)
Roles	**Roles**	Administration	• Query / Insert / Update / Delete per screen instead of on/off toggles; platform special rights (impersonate, reset client password, activate client); active role needs a screen; role with users protected	Screen registry
Admin Users	**Admin users**	Administration	• Lock / unlock, password expiry, password policy and confirmation; privileges inherited from role (no override) kept; main administrator protected	Roles
Countries	**Countries**	Master data	• Arabic name; dial-code format validation; deletion blocked when governorates reference the country	—

Old page	New page	Module	Enhancements	Dependencies
Governorates	**Governorates**	Master data	• Arabic name; same fields	Countries
Clinical Groups	**Clinical groups**	Master data	• Unit code, report template and display order	Report Templates
Services	**Services**	Master data	• Same extended editor as the client (types, sample/condition, header, aliases, precaution, members with order/sum/visibility, print-alone)	Tests Sample Types
Tests	**Tests**	Master data	• Same extended editor as the client (range bands, panic/reportable, lists, formula check, timing, frequency, delta, labels)	Clinical Groups Sample Types Units Patient Conditions
Service Global Sequence	**Service sequence**	Master data	• Unchanged behaviour, new shell	Services
Sample Types	**Sample types**	Master data	• Sample conditions per type	—
Units of Measurement	**Units of measurement**	Master data	• Reference check before deletion	—
Preparations	**Preparations**	Master data	• Option-list validation (≥ 2), Arabic question, linked services	Services
Comments	**Comment library**	Master data	• Group scope and Arabic text	Clinical Groups
Drug Types / Antibiotic Commercial Names / Antibiotics / Organisms	**Microbiology group**	Microbiology	• Moved into their own menu group; Arabic support; sensitivity values added beside them	—

3.2 New pages

New page	Module	Description	Dependencies
Subscriptions & wallets	Administration	Plans (fee, branches, users, SMS quota, analysers), client wallet balance and transactions, top-up; defines what the client-portal wallet means.	Clients, Plans
Sessions	Administration	Active admin sessions with IP and start time; terminate.	Admin Users
Audit log	Administration	Platform audit trail (client activation, master-data publication, top-ups, resets).	All administration pages
Patient conditions	Master data	Global list of patient conditions for reference-range bands.	Tests
Reasons	Master data	Global rejection, cancellation, discount and refund reasons.	—
Diagnoses	Master data	ICD codes with linked services.	Services, Code Systems (ICD-10)
Additional services	Master data	Global chargeable non-test services.	—
Report templates	Master data	Template catalogue (results, culture, text, claim, receipt, label, job order).	—
Test rules	Master data	Global auto-add rule templates.	Services
Shortcut sets	Master data	Global shortcut sets.	Services
Sensitivity values	Microbiology	S / I / R values for culture entry.	—
Analyser models	Integration library	Vendor / model library with discipline, protocol, link, modes and driver version — the catalogue tenants choose from.	—
PCR kits	Integration library	Kit interpretation library (thresholds, verdicts, folders) published to tenants.	Analyser Models
Code systems	Integration library	LOINC, CPT, ICD-10, SNOMED versions and import.	—

New page	Module	Description	Dependencies
Rule templates	Integration library	Reusable medical-rule patterns offered in the tenant rule editor.	Medical Rules (client)
Platform parameters	Platform	OTP length and validity, password minimum, session timeout, trial days, default currency and language, master-data update policy, default delay thresholds and discount cap.	—
Notification templates	Platform	SMS / email templates for OTP, portal URL, result-ready and panic notifications in both languages.	—
Profile	Platform	Administrator details and credential change.	Admin Users

3.3 New modules

New module	Description	Pages	Dependencies
Subscriptions & wallets	Plans, wallet balances, top-ups and transactions per client.	Subscriptions & wallets, Dashboard (wallet KPI)	Clients
Monitoring & governance	Sessions and platform audit log.	Sessions, Audit log	Admin Users
Extended master data	Patient conditions, reasons, diagnoses, additional services, report templates, test rules, shortcut sets, sensitivity values.	Patient conditions, Reasons, Diagnoses, Additional services, Report templates, Test rules, Shortcut sets, Sensitivity values	Services & Tests
Integration library	Analyser models, PCR kits, code systems and rule templates maintained once for all tenants.	Analyser models, PCR kits, Code systems, Rule templates	—
Platform configuration	Platform parameters and notification templates.	Platform parameters, Notification templates	—

4. Dependency map — what to set up first

1. **Platform (Super Admin):** countries and governorates → plans → clients (master-data copy generated) → integration library, code systems, rule templates, notification templates, platform parameters.
2. **Tenant foundation:** branches → roles → users (treasury requests per device) → parameters.
3. **Catalogue:** sample types and conditions → units → clinical groups and report templates → tests → services (members) → patient conditions, preparations, comment library, diagnoses, shortcut sets, test rules, reasons, additional services, doctors, sending labs.
4. **Pricing:** base price lists → payers with tiers → contract pricing (apply).
5. **Integration & automation:** analysers → analyser tests and sample codes → site agent → PCR kits → verification settings → medical rules.
6. **Operations:** registration (walk-in, home, batch, lab-to-lab) → samples → worklist / results / work queue → printing → payments, refunds, transfers, cancellations, cash closing, claims and reports.